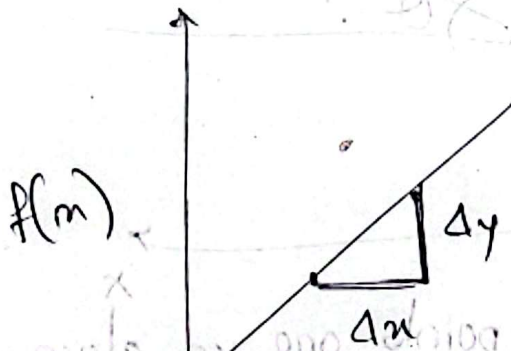


bangladesh
P. M.

Chapten 03

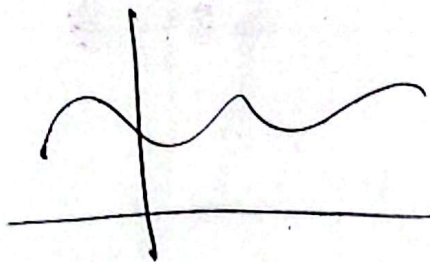
we want to find differenciation numerically.
(how fast a function changing).



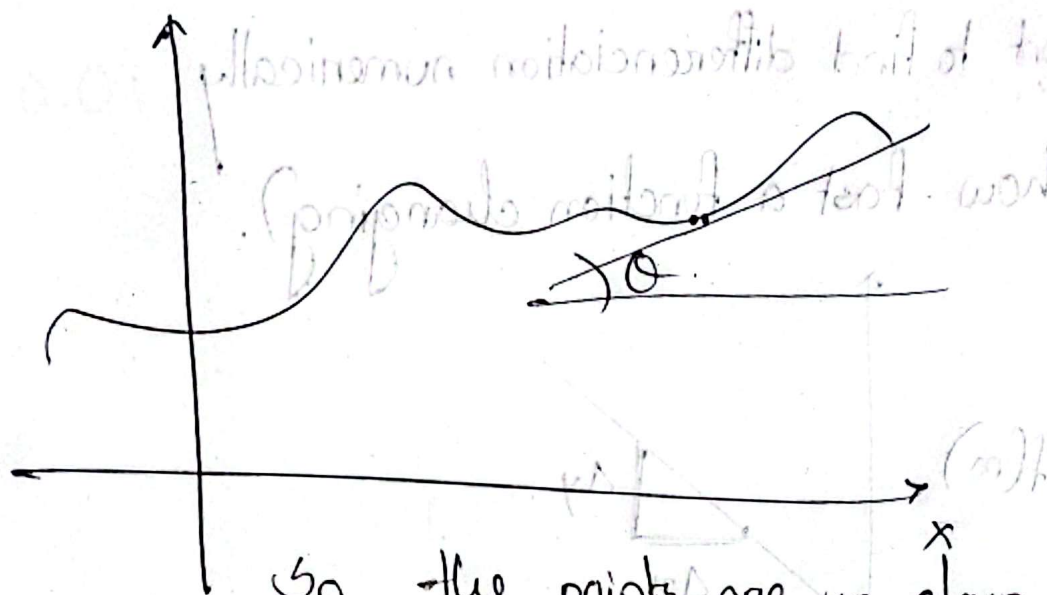
$$\frac{\Delta y}{\Delta n} = f'(n)$$

$$\tan \theta = f'(n)$$

for st. line taking 2 points give same slope.



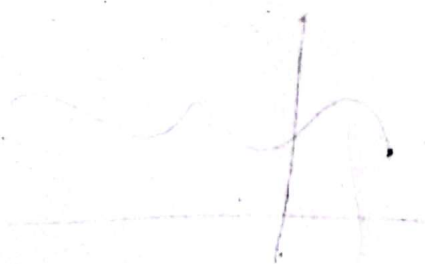
For curve it is not true, rate of change dr a



So, the points are so close that, they can work as a line. there we can create a small triangle.

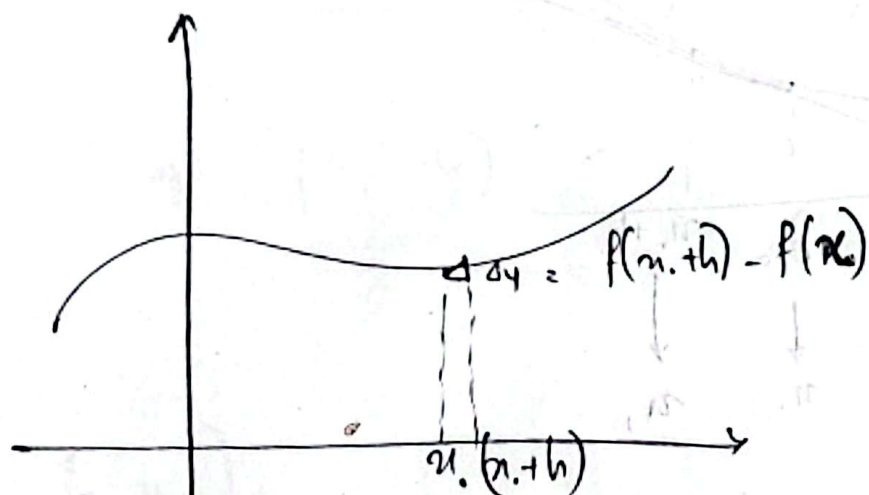
So we find a tangent at that point.

$$\text{slope} = \tan \theta = \frac{\Delta y}{\Delta x}$$



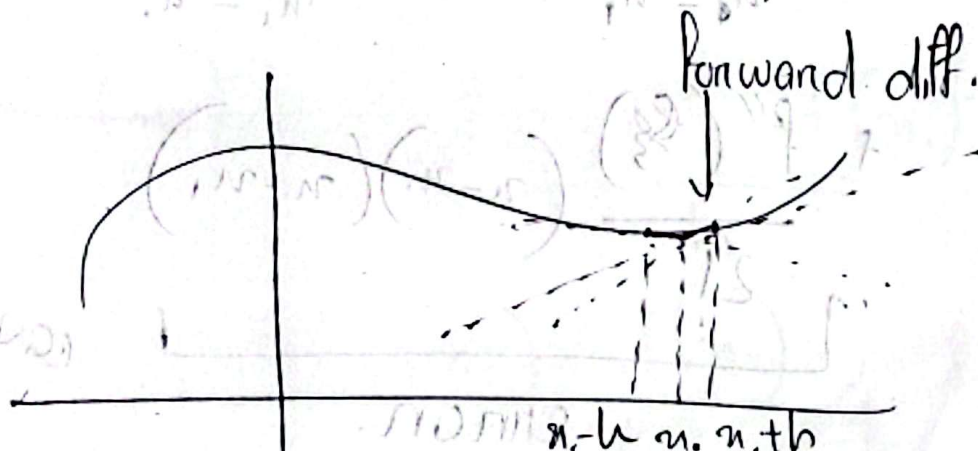
$$f'(x_1) = \lim_{h \rightarrow 0} \frac{f(x_1 + h) - f(x_1)}{h}$$

— (i)



in (i), if $h > 0$, forward diff.

if $h < 0$, backward diff.



central diff,

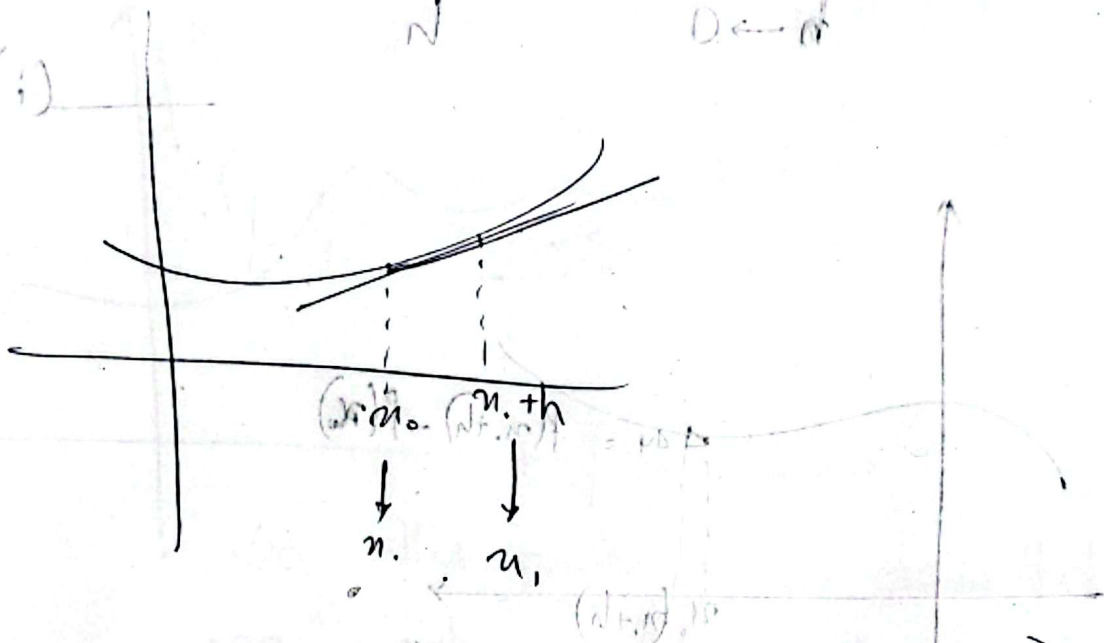
$$f'(x_1) = \frac{f(x_1 + h) - f(x_1 - h)}{2h}$$

FD:

$$\frac{(n)! - (n+1)!}{N}$$

$$\text{and } = (n)! \}$$

(i)



$$P_1(n) = \frac{n-n_1}{n_0-n_1} f(n_0) + \frac{n-n_0}{n_1-n_0} f(n_1)$$

$$f(n) = \frac{n-n_1}{n_0-n_1} f(n_0) + \frac{n-n_0}{n_1-n_0} f(n_1)$$

$$+ \frac{f''(\xi)}{2!} (n-n_0)(n-n_1)$$

Cauchy's theorem

$$\frac{(n-1)! - (n+1)!}{N} \in [n_0, n_0+h]$$

since $n - n_1 \rightarrow 0$ is the only part not constant

$$\frac{d}{dx} uv = u \frac{dv}{dx} + v \frac{du}{dx}$$

$$f'(n) = \frac{1}{n - n_1} f(n) + \frac{1}{n_1 - n} f(n_1)$$

$$+ \frac{f'''(\xi)}{2} (n - n_1)(n - n_1) \frac{d}{dx}$$

$$+ \frac{f''(\xi)}{2} (2n - n_1 - n_1)$$

$$f'(n) = \frac{1}{n - n_1} f(n) + \frac{1}{n_1 - n} f(n_1)$$

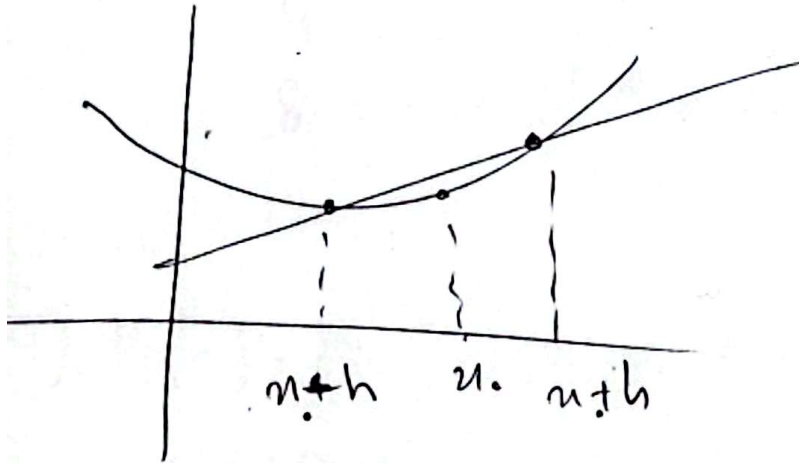
$$+ \frac{f''(\xi)}{2} (n - n_1)$$

$$\frac{f'(n_1) - f(n_1 + h) - f(n_1)}{h}$$

$$= \frac{f(n_1) - f(n_1 + h)}{n_1 - n_1 - h} + \frac{f''(\xi)}{2} (n_1 - n_1 - h)$$

$$f'(n_1) = \frac{f(n_1 + h) - f(n_1)}{h} + \frac{f''(\xi)}{2} (-h)$$

Central Difference.



$$x, \quad x_1 = x+h, \quad x_2 = \cancel{x-h} \quad \cancel{x-h} \quad x-h \quad x+2h.$$

$$f(x) = l_0(x) f(x_1) + l_1(x) f(x_2) + l_2(x) f(x_2)$$

$$= \frac{(x-x_1)(x-x_2)}{(x_1-x_1)(x_1-x_2)} f(x_1) + \frac{(x-x_1)(x-x_2)}{(x_1-x_1)(x_1-x_2)} f(x_1)$$

$$+ \frac{(x-x_1)(x-x_2)}{(x_2-x_1)(x_2-x_1)} f(x_2) + \frac{f^{(3)}(\xi)}{3!} \frac{(x-x_1)(x-x_2)}{(x-x_2)}$$

$$\begin{aligned}
 f'(x) = & \frac{2x - x_1 - x_2}{(x_0 - x_1)(x_0 - x_2)} f(x_0) + \frac{2x - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)} f(x_1) \\
 & + \frac{2x - x_0 - x_1}{(x_2 - x_0)(x_2 - x_1)} f(x_2) + \frac{f^3(x_3)}{3!} \left[(x - x_1)(x - x_2) \right. \\
 & \left. + (x - x_0)(x - x_2) + (x - x_1)(x_0 - x_2) \right] \\
 & + \frac{f^4(x_3)}{3!} (x - x_0)(x - x_1)(x - x_2)
 \end{aligned}$$

$$\begin{aligned}
 f'(x_1) = & \frac{x_1 - x_2}{(x_0 - x_1)(x_1 - x_2)} f(x_0) + \frac{2x_1 - x_0 - x_2}{(x_1 - x_0)(x_1 - x_2)} f(x_1) \\
 & + \frac{x_1 - x_0}{(x_2 - x_0)(x_2 - x_1)} f(x_2) + \frac{f^3(x_3)}{3!} (x_1 - x_0)(x_1 - x_2) \\
 = & \frac{x_0 + h - x_1 - 2h}{(-h)(-2h)} f(x_0) + \frac{2x_0 + 2h - x_1 - x_1 - 2h}{(x_1 - x_0)(x_1 - x_2)} f(x_1) \\
 & + \frac{h}{(2h)(h)} f(x_2) + \frac{f^3(x_3)}{3!} (h)(-h)
 \end{aligned}$$

$$= \frac{1}{2h} f(x_i) + \frac{1}{2h} f(x_i + 2h) - \frac{f'(x_i)}{3!} h^2$$

$$= \frac{f(x_i + 2h) - f(x_i)}{2h} - \frac{f'(x_i)}{3!} h^2$$

$$\frac{f(x_i + 2h) - f(x_i)}{2h} - \frac{f'(x_i)}{3!} h^2$$

$$\frac{f(x_i + 2h) - f(x_i)}{2h} - \frac{f'(x_i)}{3!} h^2$$

$$\frac{f(x_i + 2h) - f(x_i)}{2h} - \frac{f'(x_i)}{3!} h^2$$

$$\frac{f(x_i + 2h) - f(x_i)}{2h} - \frac{f'(x_i)}{3!} h^2$$

$$\frac{f(x_i + 2h) - f(x_i)}{2h} - \frac{f'(x_i)}{3!} h^2$$

Rounding Error

$$\text{FD, } f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

$$\text{CD, } f'(x) = \frac{f(x+h) - f(x-h)}{2h}$$

During chap -1, we saw if we subtract two close numbers then there is a big rounding error.

So in computer $f(x+h) - f(x-h)$ for $h \rightarrow 0$ gives rounding error.

$$fl[f(x_i+h)] = (1 + \delta_1) f(x_i+h)$$

$$= f(x_i+h) + \delta_1 f(x_i+h)$$

total error.

$$fl[f(x_i-h)] = (1 + \delta_2) f(x_i-h)$$

$$|\delta_1|, |\delta_2| \leq \epsilon_M,$$

$$\left| \frac{f'(x_i) \approx \frac{fl[f(x_i+h)] - fl[f(x_i-h)]}{2h}} \right|$$

$$|5 + (-1)| \leq |5| + |-1|$$

$$|4| \leq 5 + 1$$

$$|4| \leq 6$$

$$= \left| \frac{f(x_1+h) - f(x_1-h)}{2h} - \frac{f'''(\xi)}{3!} h^2 \right|$$

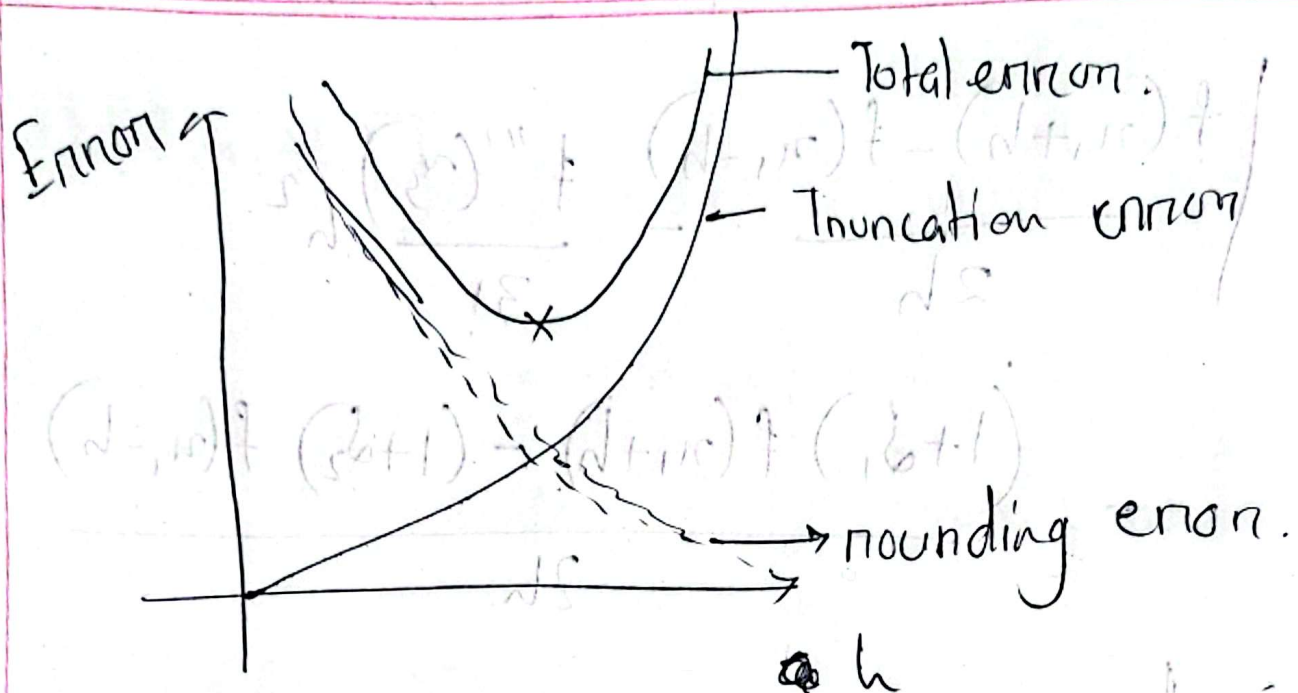
$$= \left| \frac{(1+\delta_1)f(x_1+h) - (1+\delta_2)f(x_1-h)}{2h} - \frac{f'''(\xi)}{3!} h^2 \right|$$

$$= \left| -\frac{f'''(\xi)}{3!} h^2 - \frac{\delta_1 f(x_1+h) - \delta_2 f(x_1-h)}{2h} \right|$$

$$= \left| \frac{f'''(\xi)}{3!} h^2 - \delta_1 f(x_1+h) \right| \quad \left| \begin{matrix} |a+b| \leq |a| + |b| \\ |a-b| \leq |a| + |b| \end{matrix} \right.$$

$$\left| -\frac{\delta_2 f(x_1-h)}{2h} \right| \leq \left| \frac{f'''(\xi)}{3!} h^2 \right| + \left| \frac{\delta_1 f(x_1+h) - \delta_2 f(x_1-h)}{2h} \right|$$

$$\leq \underbrace{\frac{f'''(\xi)}{3!} h^2}_{\text{error}} + \underbrace{\frac{f(x_1+h) + f(x_1-h)}{2h}}_{\text{approximation error}}$$



$$\begin{aligned}
 & \frac{(n+1) \epsilon^{\frac{1}{n+1}}}{\sqrt{h}} - \frac{(n) \epsilon^{\frac{1}{n}}}{\sqrt{h}} = \frac{(n) \epsilon^{\frac{1}{n}}}{\sqrt{h}} \left(\frac{(n+1) \epsilon^{\frac{1}{n+1}}}{(n) \epsilon^{\frac{1}{n}}} - 1 \right) \\
 & \frac{(n+1) \epsilon^{\frac{1}{n+1}}}{\sqrt{h}} - \frac{(n) \epsilon^{\frac{1}{n}}}{\sqrt{h}} \geq \frac{(n) \epsilon^{\frac{1}{n}}}{\sqrt{h}} \left(\frac{(n+1) \epsilon^{\frac{1}{n+1}}}{(n) \epsilon^{\frac{1}{n}}} - 1 \right) \\
 & \frac{(n+1) \epsilon^{\frac{1}{n+1}}}{\sqrt{h}} - \frac{(n) \epsilon^{\frac{1}{n}}}{\sqrt{h}} \geq \frac{(n) \epsilon^{\frac{1}{n}}}{\sqrt{h}} \left(\frac{(n+1) \epsilon^{\frac{1}{n+1}}}{(n) \epsilon^{\frac{1}{n}}} - 1 \right)
 \end{aligned}$$

Richardson Extrapolation.

$$D_u = \frac{f(n_1+h) - f(n_1-h)}{2h}$$

~~Ques~~ Taylor series:

$$f(n) = f(n_1) + f'(n_1)(n-n_1) + \frac{f''(n_1)}{2!}(n-n_1)^2 + \dots$$

~~$$f(n_1) = f(n_1) + f'(n_1)(n-n_1) + \frac{f''(n_1)}{2!}(n-n_1)^2 + \dots$$~~

~~$$f(n_1+h) = f(n_1) + f'(n_1)(n_1+h-n_1) + \dots$$~~

centering at n_1 :

$$f(n) = f(n_1) + f'(n_1)(n-n_1) + \frac{f''(n_1)}{2!}(n-n_1)^2 + \dots$$

$$\therefore f(n_1+h) = f(n_1) + f'(n_1)(n_1+h-n_1) + \frac{f''(n_1)}{2!}(n_1+h-n_1)^2 + \dots$$

$$+ \frac{f'''(n_1)}{3!}(n_1+h-n_1)^3 + \frac{f^{(4)}(n_1)}{4!}(n_1+h-n_1)^4 + \dots + O(h^6)$$

$$= f(u_1) + f'(u_1)h + \frac{f''(u_1)}{2!}h^2 + \frac{f'''(u_1)}{3!}h^3 + \frac{f^{(4)}(u_1)}{4!}h^4 + \frac{f^{(5)}(u_1)}{5!}h^5 + O(h^6)$$

(i)

$$\therefore f(u_1-h) = f(u_1) + f'(u_1)(u_1-h-u_1) + \frac{f''(u_1)(u_1-h-u_1)^2}{2!} + \frac{f'''(u_1)(u_1-h-u_1)^3}{3!} + \frac{f^{(4)}(u_1)(u_1-h-u_1)^4}{4!} + \frac{f^{(5)}(u_1)(u_1-h-u_1)^5}{5!} + O(h^6)$$

$$\therefore = f(u_1) - f'(u_1)h + \frac{f''(u_1)}{2!}h^2 - \frac{f'''(u_1)}{3!}h^3 + \frac{f^{(4)}(u_1)}{4!}h^4 - \frac{f^{(5)}(u_1)}{5!}h^5 + O(h^6) \quad \text{ii)}$$

$$D_n^{\phi} = \frac{1}{2h} (i - ii)$$

$$= \frac{1}{2h} \left\{ 2f'(n_1)h + \frac{2f^3(n_1)}{3!}h^3 + \frac{2f^5(n_1)}{5!}h^5 \right.$$

$$\left. + O(h^7) \right\}$$

$$= \underbrace{f'(n_1)}_{\text{exact value}} + \underbrace{\frac{f^3(n_1)}{3!}h^2 + \frac{f^5(n_1)}{5!}h^4 + O(h^6)}_{\text{error.}}$$

$$\text{error} \propto h^2$$

How to make it better?

Take combination in such a way that h^2 goes away.

Now,

$$D_h = f'(n_1) + \frac{f^3(n_1)}{3!} h^2 + \frac{f^5(n_1)}{5!} h^4 + O(h^6)$$

$$D_{h/2} = f'(n_1) + \frac{f^3(n_1)}{3!} \left(\frac{h}{2}\right)^2 + \frac{f^5(n_1)}{5!} \left(\frac{h}{2}\right)^4 + O(h^6)$$

Now, we can remove h^2 term ~~if~~.

$$\underline{2^2 D_{h/2} - D_h} = \cancel{f'(n_1)} 2^2 f'(n_1) - f'(n_1)$$

$$+ 2^2 \frac{f^5(n_1)}{5!} \left(\frac{h}{2}\right)^4 - \frac{f^5(n_1)}{5!} h^4$$

$$+ 2^2 O(h/2)^6 - O(h^6)$$

$$= \cancel{f(0)} (2^2 - 1) f'(n_1) + \left(\frac{1}{2^2} - 1 \right) \frac{f^{(5)}(n_1)}{5!} h^4 + O(h^6)$$

$$2^2 \frac{D_{h/2} - D_h}{2^2 - 1} = f'(n_1) + \frac{\left(\frac{1}{2^2} - 1 \right)}{(2^2 - 1)} \frac{f^{(5)}(n_1)}{5!} h^4 + O(h^6)$$

$$D_h = f'(n_1) + c h^n + O(h^{n+1})$$

└──────────→ can be anything

$$D_{h/2} = f'(n_1) + c \left(\frac{h}{2} \right)^n + O(h^{n+1})$$

multiply by $D_{h/2}$ with 2^n

$$D_h^{(n)} = \frac{2^n (D_{h/2}) - D_h}{2^n - 1}$$